Factor base → degree of regularity: yield and correlations on the ECC2K-130 family

2026-10-14. Study directory `research/factor_base_dor_campaign_20261013/`; preregistered hypotheses in `PREREGISTRATION.md`; run report in `reports/run-20261014T0600Z/RUN_REPORT.md`. Class: stage diagnostic of the decomposition oracle (`D*` and work-per-relation proxies); no end-to-end speedup is claimed. Curve family: Koblitz $y^2 + xy = x^3 + 1$ over F_{2^n}, toy degrees $n = 9..15$ plus $n = 131$ structural checks; the challenge member is $n = 131$ (ECC2K-130). Engine: `sage/factor_base_dor.sage`, selftest clean.

The question

Does the factor base — subspace dimension n', algebraic structure, and Shantz–Teske slack δ — change the degree of regularity D^* (Macaulay refutation degree of the Weil-descended Semaev system on non-decomposable targets) and the decomposable yield enough to point at a drastic index-calculus speedup at $n = 131$?

Findings (measured)

F1 — multiplicative closure is the only structure that lowers D^*. At matched (n, n') the subfield base (closed under products) refutes at $D^* = 2$ exactly, every seed, every cell (n=9 and n=15, both dims), against 2.35–3.00 for standard/random; in the symmetric-S_4 m=3 encoding at $n = 15, l = 3$ the subfield base refutes at $D^* = 3$ with 9 variables against $D^* = 5$ with 15 variables for standard and for the Galbraith–Gebregiyorgis geometric special subspace — two degrees, i.e. roughly two orders of magnitude of Macaulay work at fixed nvars. Normal-basis windows and symmetry breaking (`nb_window`, `nb_break`) do **not** lower D^* (they track random within ±0.2 at every matched cell).

F2 — D^* is set by the base dimension, not the field degree. Pearson $r(D^*, n') = 0.87$ pooled over the 45 non-subfield m=2 aggregates; $r(D^*, n) = -0.04$. D^* rises about one degree per $+2$ of n' across n=11 and n=13 at every structure.

F3 — the partial-closure gradient is visible below the subfield threshold. The standard base $\theta \cdot \mathrm{span}\left\{1, z, ..., z^{n'-1}\right\}$ (products stay mostly inside $\mathrm{span}\left\{1..z^{2n'}\right\}$, partially inside the base) refutes measurably lower than a random base of the same dimension at every matched cell: 2.23 vs 2.79 (n=13, $n' = 4$), 2.57 vs 2.88 (n=11, $n' = 4$), 2.86 vs 3.00 (n=13, $n' = 5$), 2.36 vs 2.99 (n=15, $n' = 5$). Consistent across all three seeds.

F4 — work per relation has an interior optimum in n'. With W\/rel = "Macaulay rows x cols at " D^* / "decomposable fraction", the optimum sits at small dimensions and the largest cells are 20–300x worse: n=13 standard $n' = 3$: 1.6×10^4 against $n' = 6$: 4.5×10^6 (286x); n=11 standard $n' = 4$: 1.0×10^5 against $n' = 5$: 4.0×10^5; n=9 subfield $n' = 3$: 7.8×10^3 against random $n' = 3$: 2.7×10^4. Yield grows $4x$ per $+1$ of n' (counting floor $|V|^2/2^n$), but D^* grows one degree per $+2$, and the Macaulay size at degree D grows combinatorially with $2n'$ — the product is minimized around $n' \approx n/3..n/4$ at toy scale.

F5 — Shantz–Teske δ buys 6–11x per relation at toy scale. Sweeping δ (base shrunk so $mn' = n - \delta$) on the standard m=2 base: work per relation falls $7.9 \times 10^5 \to 1.2 \times 10^5$ (6.5x, n=11) and $5.4 \times 10^6 \to 4.9 \times 10^5$ (11x, n=13) from $\delta = -1$ to $\delta = +3$, while D^* falls 4.00 ⟶ 2.65 and 4.00 ⟶ 2.87. Yield falls by $2^{-\delta}$ as the model predicts, and the net still favours the shrink at these sizes.

F6 — the symmetric S_4 encoding does not beat the chain on degree, only closure does. Geometric == standard at both cells ($D^* = 4$ at n=9 l=2; $D^* = 5$ at n=15 l=3); the subfield base is the only one at $D^* = 3$. The special subspace buys variables (9 vs 15), not degree — confirming the Galbraith–Gebregiyorgis caveat at the next size up.

F7 — $n = 131$ applicability. The Frobenius-window fold was verified at the challenge degree itself: a normal element exists, $\pi^{\dim V_i} = V_{i+1}$ holds, the windows are disjoint (`raw/frobwin_n131.log`). Weight-class invariant bases against windows at n=11/13 cut trials by 10x and 38x respectively (stacking tables) — an unknowns/relation-count lever, not a D^* lever. But $x^{131} - 1 = (x + 1) \cdot (\text{irreducible of degree } 130)$: **there is no proper subfield at n=131, so the one structure that measurably crushes D^* (F1) is unavailable on the challenge curve itself.**

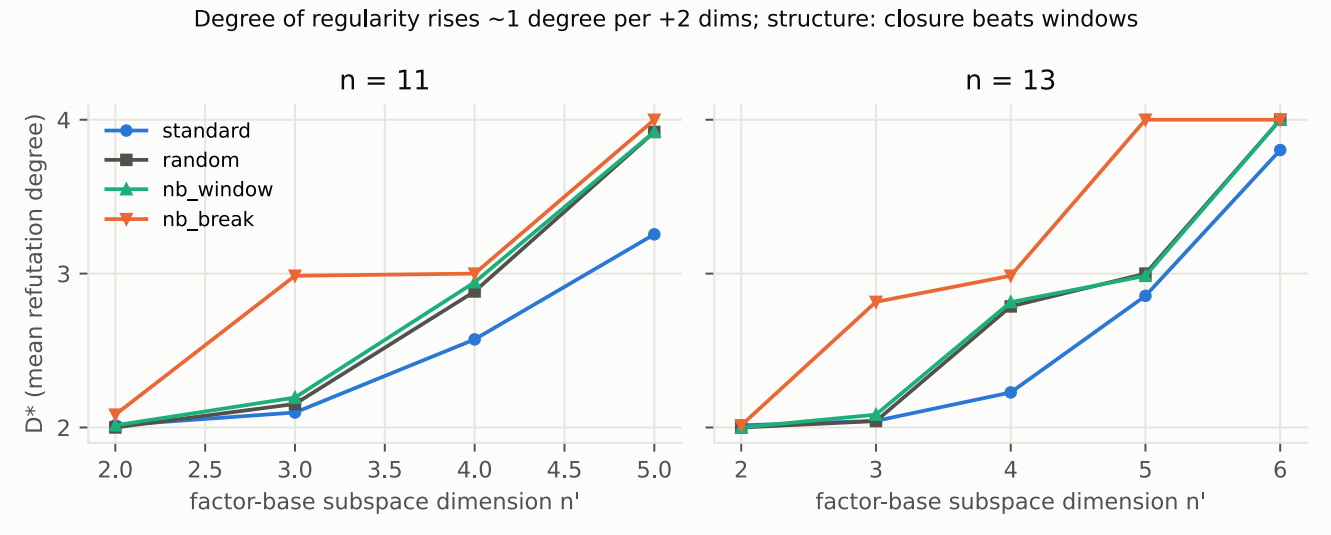


Figure 1. Mean refutation degree D^* against factor-base dimension n' (3 seeds, 24 targets/cell, m=2). Left: n=11; right: n=13. Structures: standard, random, nb_window, nb_break.

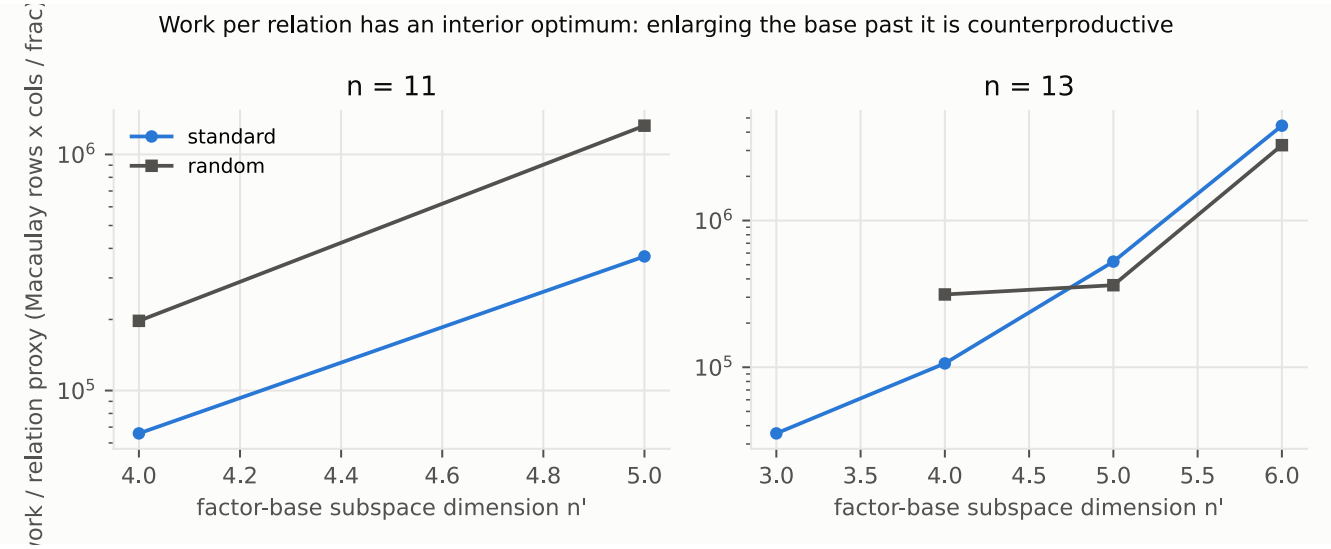


Figure 2. Work-per-relation proxy against n' (log scale). The optimum is interior; the largest bases are 20–300x worse per relation than the optimum.

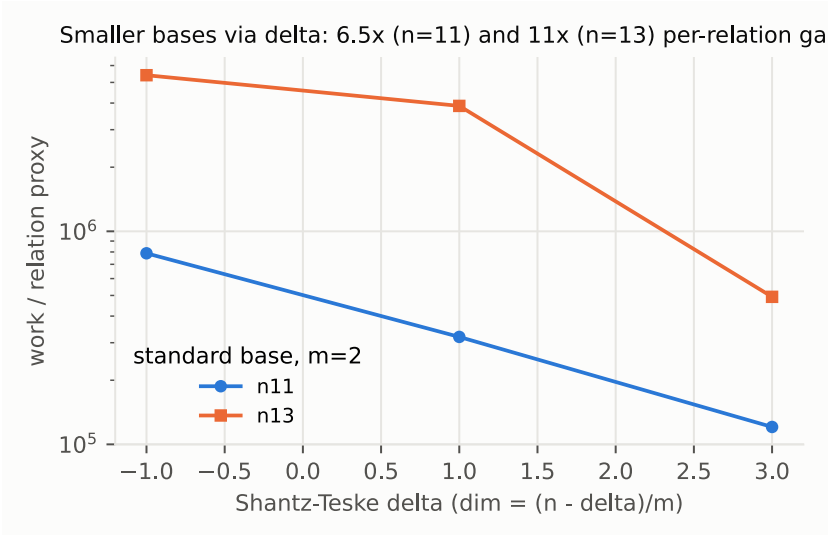


Figure 3. Shantz–Teske δ sweep, standard base, m=2. Per-relation work falls 6.5x (n=11) and 11x (n=13) from $\delta = -1$ to $\delta = +3$.

What this says about a drastic ECC2K-130 speedup

Every measured degree lever (F1, F3, F6) runs through multiplicative closure of the base, and none of the closure-free structures (windows, geometric special subspaces, weight classes) lower D^*. At $n = 131$ no proper subfield exists, so on the challenge

curve the Gröbner-oracle track's per-system cost can only be trimmed by **size** levers (F4, F5: smaller effective bases, interior optimum near $n' \approx n/3$, $\delta \approx n/3$) worth 10–300x per relation at toy scale, and by **arity** levers (the m=5/6-sum and 6-sum yield models of `RESEARCH_ECC2K130_IC_FEASIBILITY.md` §4), while the Frobenius-invariant weight-class bases (F7) cut the number of systems by n with the windows verified at 131. The counting floor (yield $\approx |V|^m/2^n$) fixes how small a base can be before relations vanish; at $n = 131$ that floor is what forces $K \geq 2 \times 10^7$ columns in the 4-sum ladder, so the size levers trade against it and must be re-priced at m=83 under the repository's confidence gate before any transfer claim.

Classification and falsification record

H1 confirmed (closure lowers D^*; windows do not). H2 confirmed ($r = 0.87$ with n', interior optimum of $\mathrm{W}\backslash/\mathrm{rel}$). H3 confirmed (6.5–11x at $\delta = +3$; the $2^{-\delta}$ yield cost is visible in the recorded fractions). H4 confirmed (geometric = standard on D^*). H5 confirmed ($r(D^*, n) = -0.04$). Chained m=3 re-measurement: not completed (engine budget, preserved failures); prior $D^* = 5$ figure stands as prior evidence. All numbers are toy-scale stage diagnostics; nothing here is an end-to-end or sub-ρ claim, and the m=83 gate (AGENTS.md §8a) is unmet by this study.